Interactive Machine Teaching by Labeling Rules and Instances

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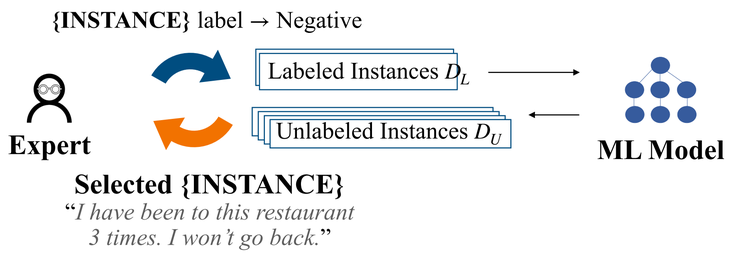
Data labeling is expensive

Approach 1: Active Learning

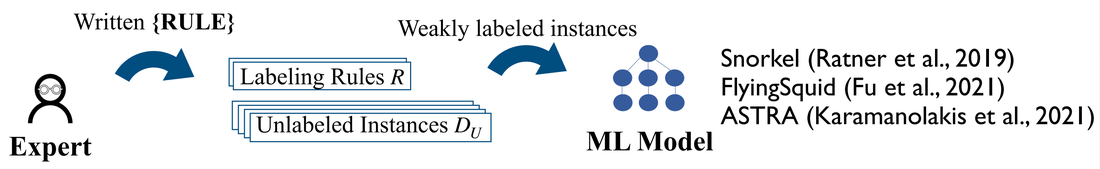
Adaptively choose which instances to label



Issue: Labeling one instance at a time is not scalable.

Approach 2: Weak Supervision

Collect expert rules that automatically create many weak labels



Rule examples:

Spam classification

```
def regex_check_out(x):
    return SPAM if re.search("check.*out", x) else ABSTAIN
```

Question type classification

```
def numeric_question(x):
    return NUMERIC if x.startswith("when") else ABSTAIN
```

Issue: Designing many strong rules in a single shot is hard.

How to efficiently use expert feedback?

Assume limited budget T to query an expert

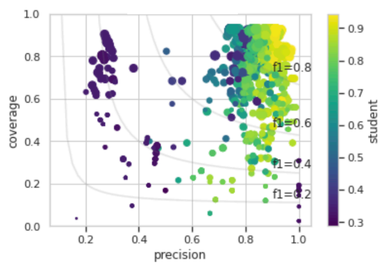
- Should we spend T querying for **rules** or **instance labels**?
- What **rule properties** are required to train an accurate model?

Our Work

1. Prevalent patterns in Weak Supervision

We unify weak supervision methods via a Teacher-Student abstraction::

- Teacher: uses rules to weakly label D_U
- Student: trained with soft Teacher labels



We evaluate 1,000+ Teacher-Student pairs:

- Higher Teacher $F1 \neq$ higher Student $F1$
- Teacher precision is more important than coverage

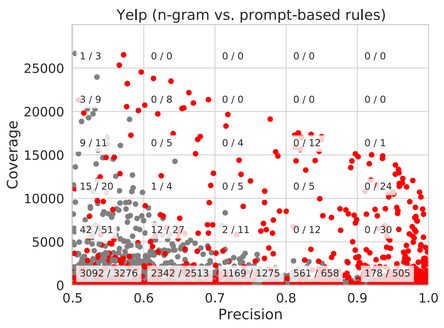
2. Automatic rule extraction via prompting

We propose a method that extracts rules with rich predicates:

- n-gram features
- Syntactic features
- Prompt-based features

Candidate Rules (predicate → label)
PMT-EXPERIENCE="terrible" → Negative
PMT-EXPERIENCE="fantastic"→ Positive
PMT-RECOMMEND="certainly" → Positive
PMT-IS_ABOUT="prizes" → Spam
NGRAM="http" **AND** PMT-ASKS_FOR="donations" → Spam
NER="CARDINAL" **AND** PMT-ASKS_FOR="information"→ Spam

Our rule family achieves higher precision and coverage than n-gram rules.



3. Interactive machine teaching

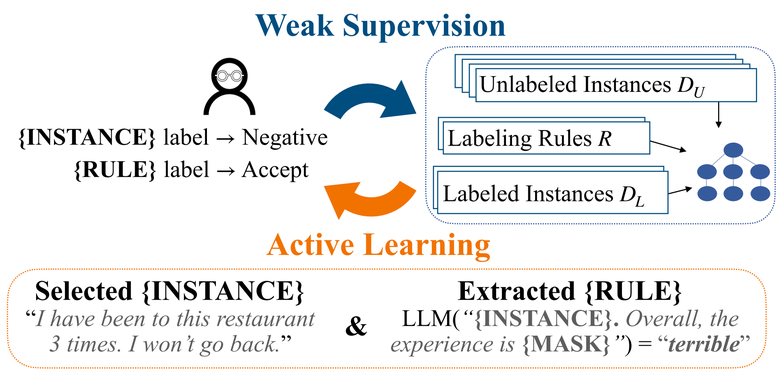
We present a human-in-the-loop machine teaching framework, which queries for expert feedback on both instances and rules.

INTERVAL: Interactive Learning with Weak Supervision

Balances the quality of **instance labels** with the efficiency of **labeling rules**

Asks expert feedback on both instances and rules

Automatically extracts labeling rules



Algorithm

Initialize $D'_L = D_L$, $R' = R$
Repeat until the budget T runs out:
3.1: Train Teacher $q^*_\phi(\cdot)$ and Student $p_\theta(\cdot)$ using D'_L, D_U, R'
3.2: Apply $p_\theta(\cdot)$ to $s \in D_U$ to obtain soft labels: $D_{Student} = \{(s_i, \mathbf{p}_i)\}_{s_i \in D_U}$
3.3: Pick a candidate instance $s_i \in D_U$
3.4: Query the label y_i for s_i (cost $= T_I$)
3.5: Extract candidate rules r^j that cover s_i
3.6: Query the labels z^j for β_i rules r^j (cost $= \beta_i \cdot T_R$)
3.7: Update $D'_L = D'_L \cup \{(s_i, y_i)\}_{\beta_i}$, $R' = R' \cup \{r^j : (v^j(\cdot), z^j)\}$, $T = T - T_I - \beta_i \cdot T_R$

Example

Text instance s_i:
"Prime Minister Manmohan Singh today said international environment for India's development was highly favourable..."
Queries:
- Instance label: World
- Rule 1: NGRAM="prime minister" → World ✓
- Rule 2: PROMPT_IS_ABOUT="politics" → World ✓
- Rule 3: NGRAM="international" → World ✗
- Rule 4: –
- Rule 5: –

Table 11: Example from AGNews with $\beta = 5$. All classes are "World," "Sports," "Business," and "Sci/Tech." Out of the rules that were queried, 2 were accepted and 1 was rejected.

INTERVAL outperforms Weak Supervision and Active Learning using less resources

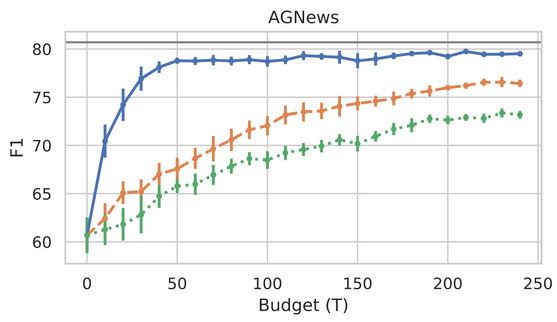
Experimental Results

INTERVAL is more effective than existing Weakly Supervised Learning (WSL) and Active learning approaches even when starting with no expert-written rules.

Method	$\lvert D_L \rvert$	D_U	R	T (T_I, T_R)	YouTube	SMS	IMDB	Yelp	TREC	AGNews	AVG F1
Fully Supervised	100%	-	-	-	94.0	95.6	79.6	87.5	90.3	80.7	88.0
Low Supervised	20·K	-	-	-	79.8	82.5	61.6	70.4	55.0	58.8	68.0
Semi Supervised	20·K	✓	-	-	80.7	83.2	63.4	72.0	55.0	60.7	69.2
WSL (ASTRA)	20·K	✓	✓	-	**90.0**	86.8	**71.2**	80.2	57.0	**75.9**	**76.8**
Active Learning (hierarchical)	20·K	✓	-	100 (100, 0)	85.3	**89.9**	67.6	**81.2**	**61.4**	71.4	76.1
INTERVAL	20·K	✓	-	100 (50, 50)	**91.4**	**94.8**	**79.3**	**86.2**	**66.6**	**78.8**	**82.8**

Efficient use of expert feedback

INTERVAL requires as few as $T = 1$ queries to reach $F1$ values that Active Learning cannot match even with $T = 100$ queries.



Rule vs. instance trade-off

Feedback on both rules and instances is more effective than feedback on instances only even when labeling rules are up to 9x more expensive than labeling instances.

Discussion & Future Work

- INTERVAL prompts pre-trained models during training only, and can work with any model for inference, enabling applications where **deploying LLMs is impossible**.
- A higher-$F1$ Teacher **does not necessarily lead** to a higher-$F1$ Student. Would this apply to other Teacher types or distillation types?
- Integrating **richer feedback** (e.g., editing rules) could lead to stronger gains.
- Could **LLMs replace experts** without sacrificing accuracy in interactive learning?

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